

Erdős Problem #563

EP-563

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $F(n, \alpha)$ denote the smallest m such that there exists a 2-colouring of the edges of K_n so that every $X \subseteq [n]$ with $|X| \geq m$ contains more than $\alpha \binom{|X|}{2}$ many edges of each colour.

Prove that, for every $0 \leq \alpha < 1/2$, $F(n, \alpha) \sim c_\alpha \log n$ for some constant c_α depending only on α .

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>